

PLANNING, INSTRUCTION, AND TECHNOLOGY

chapter outline

① Planning

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Instructional Planning

Time Frames and Planning

② Teacher-Centered Lesson Planning and Instruction

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Teacher-Centered Lesson Planning

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③ Learner-Centered Lesson Planning and Instruction

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Standards for Technology-Literate Students

Teaching, Learning, and Technology

Education is the transmission of civilization.

—Ariel and Will Durant

American Authors and Philosophers, Twentieth Century



Connecting with Teachers Lois Guest and Kevin Groves

In Lois Guest's fifth-grade classroom at Hesperian Elementary School in San Lorenzo, California, 30 students are working on their new laptop computers provided by the school district. A month after the computers arrived, outdated encyclopedias sit on shelves, and students no longer have to compete for time in the school's computer lab.

The new laptops mark a monumental shift in Guest's career. She has taught for 35 years at Hesperian Elementary and remembers when slide projectors were the height of high tech. Now, she is building a Web site for her class and learning how to use a digital camera and an optical scanner. Guest says that she is learning a lot about technology and that her students are also teaching her how to do things on the computer. Ten-year-old Bianca Gutierrez said that she enjoys working on her new laptop computer so much that it makes her look forward to coming to school.

When Kevin Groves took his fifth-grade class on a field trip to a botanical garden, his students insisted that they be allowed to bring their laptops. One of his students, Salvador Mata, sat on the grass with his mother and explained to her how to make a PowerPoint

presentation. His mother is pleased about Salvador learning how to use a computer at such an early age, believing that it will help him get a good job later in his life. Groves commented that he researches every link he puts on his Web site to make sure it is educational and won't lead to something bad for his students. Source: All-USA First Teacher Team. *USA Today*, 2001.



Lois Guest (left) and Kevin Groves (right) working with students on their PowerPoint presentations.

(Left) Mark Costantini/San Francisco Chronicle/Polaris; (Right) Kevin Groves

Preview

This chapter builds on the discussions of learning and cognitive processes in other chapters and addresses instructional planning at the level of the overall lesson plan or unit. We explore teacher-centered lesson planning, building on the behavioral principles covered in the chapter on behavioral and social cognitive approaches, and learner-centered lesson planning, building on material covered in other chapters of this text. Finally, we explore important classroom applications of technology, such as use of the Internet.

LG 1 Explain what is involved in classroom planning.

1 PLANNING

Instructional Planning

Time Frames and Planning

It has been said that when people fail to plan, they plan to fail. Many successful people attribute their accomplishments to effective planning. Our introduction to planning describes what instructional planning is and the different time frames of planning.

INSTRUCTIONAL PLANNING

Planning is a critical aspect of being a competent teacher (Burden & Byrd, 2016; Enow & Goodwyn, 2018). **Instructional planning** involves developing a systematic, organized strategy for lessons. Teachers need to decide what and how they are going to teach before they do it. Although some wonderful instructional moments are spontaneous, lessons still should be carefully planned (Borich & Blanchette, 2022).

instructional planning A systematic, organized strategy for planning lessons.

Instructional planning might be mandated by the school in which you teach. Many principals and instructional supervisors require teachers to keep written plans, and you may be asked to submit lesson plans several weeks in advance. When observing classroom teachers, supervisors check to see if the teacher is following the plan. If a teacher is absent, a substitute teacher can follow the plan.

Expectations for teacher planning have increased with the promulgation of state learning standards that specify what students need to know and be able to do (Guillaume, 2016). However, these standards usually do not state what the teacher should do in the classroom to reach these standards. When standards are in place, teachers must figure out how to plan and organize their curriculum around the most important dimensions implied by the standards and create a “sequence and set of learning activities for the particular students they teach” (Darling-Hammond et al., 2005, p. 184).

Many planning strategies are organized around four elements: “the nature of the subject matter, the learners, the context, and the teacher’s role” (Darling-Hammond et al., 2005, p. 184). One effective planning strategy that many teachers use is *mapping backward* from “goals to desired performances to activities and elements of scaffolding needed to support student progress.” Indeed, a good strategy is to begin your planning by thinking about what goals you want your students to reach by the end of the school year and then map backward from that point. In one analysis, many experienced teachers described how in the early years of their teaching they lacked a long-term vision in their curriculum planning; they strongly urged beginning teachers to think more about the big picture—the key things they want their students to learn this year—and how they can guide their students to get there (Kunzmann, 2003).

Linda Darling-Hammond and her colleagues (2005) described some important aspects of curriculum planning:

Teachers must decide what is important to include, given their goals, and know how to make it accessible to a particular group of students. This requires thinking about how to give students a schema or conceptual map of the domain to be studied (National Research Council, 2000) as well as planning specific activities in light of students’ levels of readiness for various kinds of learning experiences. It also requires consideration of the kinds of information, demonstrations, models, inquiry opportunities, discussion, and practice students need over time to understand particular concepts and develop particular skills. (pp. 185–186)

In sum, teachers need to “figure out which things students should do when, in what order, and how” to implement the big picture of their curricular vision.

TIME FRAMES AND PLANNING

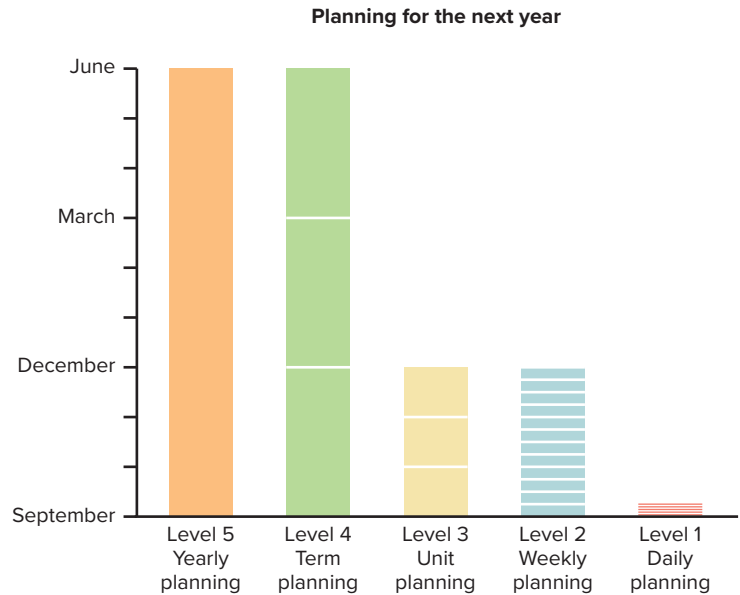
As we just indicated, developing systematic time plans involves knowing what needs to be done and when to do it, or focusing on “task” and “time.”

You will need to plan for different time spans, ranging from yearly to daily planning (Parkay, 2020). If school-wide planning or your own career planning is involved, the time frame likely will be a number of years. Robert Yinger (1980) identified five time spans of teacher planning: yearly planning, term planning, unit planning, weekly planning, and daily planning. Figure 1 illustrates these time frames and shows planning for them. Yinger also recommends that teachers attend to four areas when planning: goals, sources of information, the form of the plan, and criteria for the effectiveness of the planning.

Although planning is a key dimension of successful teaching, don’t overplan to the point of becoming an automaton. Develop organized plans and try to carry them out, but be flexible; as a year, month, week, or day unfolds, adapt to changing

FIGURE 1 FIVE TIME SPANS OF TEACHING PLANNING AND THEIR OCCURRENCE OVER THE SCHOOL YEAR

Source: "A Study of Teacher Planning," by R. J. Yinger, 1980, *The Elementary School Journal*, 80 (3), p. 113. University of Chicago Press.



How does time allotted to lesson planning differ in Chinese and American schools?
Dwight Cendrowski/Alamy Stock Photo

Thinking Back/Thinking Forward

Think of integrated assessment planning in terms of three time frames: preinstruction, during instruction, and postinstruction. Connect to "Classroom Assessment and Grading"



circumstances. A controversial current event or necessary topic might emerge that you did not originally include. Monitor and rework your plans as the school year goes by to suit these changing circumstances (Borich & Blanchette, 2022).

A final comment needs to be made about planning. It is increasingly understood that teachers need to monitor and evaluate their curriculum planning in terms of how well students are making progress toward learning goals as they go through the term. Thus, time for assessment and feedback to students needs to be built into the planning process (Brookhart & Nitko, 2019; Popham, 2020).

A final consideration involves how much time schools allot to lesson planning in the United States compared with other countries. Cross-cultural comparisons reveal that lesson planning is given a much higher priority in Chinese than American schools (Shen, Poppink, et al., 2007; Shen, Zhen, & Poppink, 2007). Many American teachers' schedules allow little time for lesson planning, in many cases 30 minutes or less, during the school day. By contrast, Chinese teachers

teach only one or two hours a day, in a core subject area, and they spend extensive time on lesson planning: on average about two hours a week of formal collaboration and two hours a week of informal collaboration with colleagues in one core subject area. They also have one to two hours a day to correct homework and classwork, and 30 minutes a day to provide homework feedback to students.

What planning strategies do teachers use to make their teaching more effective? Following are their responses.

EARLY CHILDHOOD Lesson planning for our preschoolers is done by a team of two at the beginning of the month. The team determines what will be covered by reviewing skills that need to be mastered and implementing appropriate activities. We also incorporate the children's interests in building our plans.



—VALARIE GORHAM, *Kiddie Quarters, Inc.*

ELEMENTARY SCHOOL: GRADES K-5 To prepare for my second-grade classes, I plan each week's (and sometimes two weeks') lessons the Thursday before. This way I can have all copies made, have the necessary materials, and be sure additional materials are available (e.g., library books). I then map out the week and check that time is set aside to reteach and reinforce material.



—JANINE GUIDA POUTRE, *Clinton Elementary School*

MIDDLE SCHOOL: GRADES 6-8 When planning my lessons, I use the backward design method: I make a list of essential questions that I want my students to know when they finish the activity. These essential questions go on my whiteboard so that the students are aware of what they are going to learn. By knowing exactly what I want the students to learn, I can plan a successful lesson.



—CASEY MAASS, *Edison Middle School*

HIGH SCHOOL: GRADES 9-12 I get to school by 6:00 a.m. to plan and lay out what I need for the day. I need quiet time without others around and access to copy machines without a line. By 7:00 a.m. I have my activities for the day laid out and ready to go. Of course, I am a morning person!



—SANDY SWANSON, *Menomonee Falls High School*

Review, Reflect, and Practice

1 Explain what is involved in classroom planning.

REVIEW

- What is instructional planning? Why does instruction need to be planned?
- What planning needs to be done related to the use of time?

REFLECT

- In your own K-12 experiences, did you ever have a teacher who did not put enough effort into planning? What was that like for students?

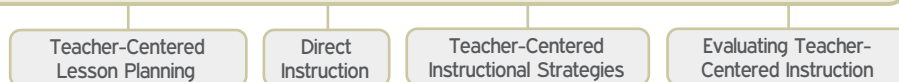
PRAXIS™ PRACTICE

- Ms. Swenson is planning for the school year. Which of the following strategies should she follow?
 - Plan forward, starting with plans for the first week of school.
 - Map backward, determining the goals she wants her students to reach by the end of the school year.
 - Not spend too much time in planning but rather get a sense of where her students are in the first month of school.
 - Engage in circular planning.
- In planning, Mr. Tomasello considers the objectives established by the district and state-mandated standards and aligns his curriculum to them. After doing this, he considers what he wants to teach when, based on the sequence that makes the most sense. Finally, he reserves the necessary equipment and requests the appropriate materials to carry out his plans. At what level of planning is Mr. Tomasello most likely to be engaging?
 - daily
 - weekly
 - term
 - yearly

Please see answer key at end of chapter

LG 2 Identify important forms of teacher-centered instruction.

2 TEACHER-CENTERED LESSON PLANNING AND INSTRUCTION



Traditionally, the focus in schools has been on teacher-centered lesson planning and instruction. In this approach, planning and instruction are highly structured and the teacher directs students' learning.

TEACHER-CENTERED LESSON PLANNING

Three general tools are especially useful in teacher-centered planning. These are behavioral objectives, task analysis, and instructional taxonomies (classifications), which we explore next.

Behavioral Objectives Behavioral objectives are statements about changes that the teacher wishes to see in students' performance. Behavioral objectives should be very specific and often have three parts (Mager, 1962):

- *Student's behavior.* Focus on what the student will learn or do.
- *Conditions under which the behavior will occur.* State how the behavior will be evaluated or tested.
- *Performance criteria.* Determine what level of performance will be acceptable.

For example, a teacher might develop a behavioral objective around the idea that the student will describe five causes of the decline of the British Empire (student's behavior). The teacher plans to give the student an essay test on this topic (conditions under which the behavior will occur). And the teacher decides that explaining four or five causes will be acceptable performance (performance criterion).

Task Analysis Another tool of teacher-centered planning is **task analysis**, which focuses on breaking down a complex task that students are to learn into its component parts (Alberto & Troutman, 2022). The analysis can proceed in three basic steps (Moyer & Dardig, 1978):

1. Determine what skills or concepts the student needs to have to learn the task.
2. List any materials that will be required in order to perform the task, such as paper, pencil, and calculator.
3. List all of the components of the task in the order in which they must be performed.

Instructional Taxonomies Instructional taxonomies also aid teacher-centered approaches. A **taxonomy** is a classification system. **Bloom's taxonomy** was developed by Benjamin Bloom and his colleagues (1956). It classifies educational objectives into three domains: cognitive, affective, and psychomotor. Bloom's taxonomy has been used by many teachers in their lesson planning to create goals and objectives (Tarman & Kuran, 2015; Yanchinda et al., 2016).

The Cognitive Domain Bloom's cognitive taxonomy has six objectives (Bloom et al., 1956):

- *Knowledge.* Students have the ability to remember information. For example, an objective might be to list or describe four main advantages of using a computer for word processing.
- *Comprehension.* Students understand the information and can explain it in their own words. For example, an objective might be to explain or discuss how a computer can effectively be used for word processing.

behavioral objectives Statements that communicate proposed changes in students' behavior to reach desired levels of performance.

task analysis Breaking down a complex task that students are to learn into its component parts.

taxonomy A classification system.

Bloom's taxonomy Developed by Benjamin Bloom and colleagues; classifies educational objectives into three domains—cognitive, affective, and psychomotor.

- *Application.* Students use knowledge to solve real-life problems. For example, an objective might be to apply what has been learned about using a computer for word processing to how this could be used in various careers.
- *Analysis.* Students break down complex information into smaller parts and relate information to other information. For example, an objective might be to compare one type of word-processing program with another for doing term papers.
- *Synthesis.* Students combine elements and create new information. For example, an objective might be to organize all that has been learned about the use of computers for writing.
- *Evaluation.* Students make good judgments and decisions. For example, an objective might be to critique different word-processing programs or to judge the strengths and weaknesses of a particular word-processing program.



What objectives are involved in Bloom's cognitive taxonomy?
Hero Images/Getty Images

The Affective Domain The affective taxonomy consists of five objectives related to emotional responses to tasks (Krathwohl et al., 1964). Each of the following five objectives requires the student to show some degree of commitment or emotional intensity:

- *Receiving.* Students become aware of or attend to something in the environment. For example, a guest comes to class to talk with students about reading. An objective might be for students to listen carefully to the speaker.
- *Responding.* Students become motivated to learn and display a new behavior as a result of an experience. An objective might be for students to become motivated to become better readers as a result of the guest speaker's appearance.
- *Valuing.* Students become involved in, or committed to, some experience. An objective might be for students to value reading as an important skill.
- *Organizing.* Students integrate a new value into an already existing set of values and give it proper priority. An objective might be to have students participate in a book club.
- *Value characterizing.* Students act in accordance with the value and are firmly committed to it. An objective might be that over the course of the school year, students increasingly value reading.

The Psychomotor Domain Most of us link motor activity with physical education and athletics, but many other subjects, such as handwriting and word processing, also involve movement. In the sciences, students have to manipulate complex equipment; the visual and manual arts require good hand-eye coordination. Bloom's psychomotor objectives include these:

- *Reflex movements.* Students respond involuntarily without conscious thought to a stimulus—for example, blinking when an object unexpectedly hurtles their way.
- *Basic fundamentals.* Students make basic voluntary movements that are directed toward a particular purpose, such as grasping a microscope knob and correctly turning it.
- *Perceptual abilities.* Students use their senses, such as seeing, hearing, or touching, to guide their skill efforts, such as watching how to hold an instrument in science, like a microscope, and listening to instructions on how to use it.
- *Physical abilities.* Students develop general skills of endurance, strength, flexibility, and agility, such as running long distances or hitting a softball.

- *Skilled movements.* Students perform complex physical skills with some degree of proficiency, such as effectively sketching a drawing.
- *Nondiscursive behaviors.* Students communicate feelings and emotions through bodily actions, such as doing pantomimes or dancing to communicate a musical piece.

Teachers can use Bloom's taxonomies for the cognitive, affective, and psychomotor domains to plan instruction. In the past, instructional planning has generally focused on cognitive or behavioral objectives. Bloom's taxonomy provides for a more expansive consideration of skills by also including affective and psychomotor domains.

A group of educational psychologists updated Bloom's knowledge and cognitive process dimensions in light of contemporary theory and research (Anderson & Krathwohl, 2001). In the update, the knowledge dimension has four categories, which lie along a continuum from concrete (factual) to abstract (metacognition):

- *Factual:* The basic elements students must know to be acquainted with a discipline or solve problems in it (technical vocabulary, sources of information)
- *Conceptual:* The interrelationships among the basic elements within a larger structure that allow them to function together (periods of geological time, forms of business ownership)
- *Procedural:* How to do something, methods of inquiry, and criteria for using skills (skills used in painting with watercolors, interviewing techniques)
- *Metacognitive:* Knowledge of cognition and awareness of one's own cognition (knowledge of outlining and strategies for remembering)

In the update of the cognitive process dimension, six categories lie along a continuum from less complex (remember) to more complex (create):

- *Remember.* Retrieve relevant knowledge from long-term memory. (Recognize the dates of important events in U.S. history.)
- *Understand.* Construct meaning from instruction that includes interpreting, exemplifying, classifying, summarizing, inferring, comparing, and explaining. (Explain the causes of important eighteenth-century events in France.)
- *Apply.* Carry out or use a procedure in a given situation. (Use a law in physics in situations in which it is appropriate.)
- *Analyze.* Break material into its component parts and determine how the parts relate to each other and to overall structure or purpose. (Distinguish between relevant and irrelevant numbers in a math word problem.)
- *Evaluate.* Make judgments based on criteria and standards. (Detect inconsistencies or fallacies in a product.)
- *Create.* Put elements together to form a coherent or functional whole; reorganize elements into a new pattern or structure. (Generate hypotheses to account for an observed phenomenon.)

direct instruction approach A structured, teacher-centered approach characterized by teacher direction and control, high teacher expectations for students' progress, maximum time spent by students on academic tasks, and efforts by the teacher to keep negative affect to a minimum.



How have the objectives been modified in the update of Bloom's cognitive taxonomy?

Mel Yates/Photodisc/Getty Images

DIRECT INSTRUCTION

Direct instruction is a structured, teacher-centered approach that is characterized by teacher direction and control, high teacher expectations for students' progress, maximum time spent by students on academic tasks, and efforts by the teacher to keep negative affect to a minimum. The focus of direct instruction is academic activity; nonacademic materials (such as toys, games, and puzzles) tend not to be used; also deemphasized is nonacademically oriented teacher-student interaction (such as conversations about self or personal concerns) (Burden & Byrd, 2016).

Teacher direction and control take place when the teacher chooses students' learning tasks, directs students' learning of the tasks, and minimizes the amount of nonacademic talk. The teacher sets high standards for performance and expects students to reach these levels of excellence.

An important goal in the direct-instruction approach is maximizing student learning time (Borich & Blanchette, 2022). Time spent by students on academic tasks in the classroom is called *academic learning time*. Learning takes time. The more academic learning time students experience, the more likely they are to learn the material and achieve high standards. The premise of direct instruction is that the best way to maximize time on academic tasks is to create a highly structured, academically oriented learning environment.

Cross-cultural research about the amount of time students spend on math in different countries, as well as other comparisons across countries, reveals that students in many Asian countries spend more time and effort in learning than do U.S. students. Harold Stevenson, one of the leading experts on children's learning, conducted research on this topic for five decades. In the 1980s and 1990s, he turned his attention to discovering ways to improve children's learning by conducting cross-cultural comparisons of children in the United States with children in Asian countries, especially Japan, China, and Taiwan (Stevenson, 1992, 1995, 2000; Stevenson & Hofer, 1999; Stevenson et al., 1990). In Stevenson's research, Asian students consistently outperformed American students in mathematics. Also, the longer students were in school, the wider the gap became—the lowest difference was in first grade, the highest in eleventh grade (the highest grade studied).

To learn more about the reasons for these cross-cultural differences, Stevenson and his colleagues spent thousands of hours observing in classrooms, as well as interviewing and surveying teachers, students, and parents. They found that Asian teachers spent more of their time teaching math than American teachers did. For example, in Japan more than one-fourth of the total classroom time in first grade was spent on math instruction, compared with only one-tenth of the time in U.S. first-grade classrooms. Also, Asian students were in school an average of 240 days a year, compared with 178 days in the United States.

Also, differences were found between the Asian and American parents. American parents had much lower expectations for the children's education and achievement than did the Asian parents. Also, the American parents were more likely to say that their children's math achievement was due to innate ability; the Asian parents were more likely to say that their children's math achievement was the consequence of effort and training. The Asian students were more likely to do math homework than the American students, and the Asian parents were far more likely to help their children with their math homework than were the American parents (Chen & Stevenson, 1989).

In addition to maximizing student learning time, another emphasis in the direct-instruction approach is keeping negative affect to a minimum. Researchers have found that negative affect interferes with learning (Merrell et al., 2007), even in a digital setting (Wortha et al., 2019). Advocates of direct instruction underscore the importance of keeping an academic focus and avoiding negative affect, such as the negative feelings that can often arise in both the teacher and students when a teacher overcriticizes (Sieberer-Nagler, 2016).

Thinking Back/Thinking Forward

A major curriculum controversy is whether teaching should mainly follow a direct instruction or constructivist approach. Connect to "Educational Psychology: A Tool for Effective Teaching."



DIVERSITY



RESEARCH



DEVELOPMENT



In Stevenson's research, Asian students scored considerably higher than U.S. students on math achievement tests. *What are some possible explanations for these findings?*

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TEACHER-CENTERED INSTRUCTIONAL STRATEGIES

Many teacher-centered strategies reflect direct instruction. Here we will talk about orienting students to new material; lecturing, explaining, and demonstrating; questioning and discussing; mastery learning; seatwork; and homework.

Orienting Before presenting and explaining new material, establish a framework for the lesson and orient students to the new material: (1) review the previous day's activities; (2) discuss the lesson's objective; (3) provide clear, explicit instructions about the work to be done; and (4) give an overview of today's lesson.

Advance organizers are teaching activities and techniques that establish a framework and orient students to material before it is presented. You can use advance organizers when you begin a lesson to help students see the "big picture" of what is to come and how information is meaningfully connected.

Advance organizers come in two forms: expository and comparative. **Expository advance organizers** provide students with new knowledge that will orient them to the upcoming lesson. The chapter-opening outline and learning goals in each chapter of this text are expository advance organizers. Another way to provide an expository advance organizer is to describe the lesson's theme and why it is important to study this topic. For example, in orienting students to the topic of exploring the Aztec civilization in a history class, the teacher says that they are going to study the Spanish invasion of Mexico and describes who the Aztecs were, what their lives were like, and their artifacts. To heighten student interest, she also says that they will study worlds in collision as Spain's conquistadors were filled with awe at sights of a spectacular Western civilization. There are Mexican American students in her class, and the teacher emphasizes how this information can help everyone in the class understand these students' personal and cultural identity.

Comparative advance organizers introduce new material by connecting it with what students already know. For example, in the history class just mentioned, the teacher says that the Spanish invasion of Mexico continued the transatlantic traffic that changed two worlds: Europe and the Americas. She asks students to think about how this discussion of the Aztecs connects with Columbus's journey, which they examined last week.



Lecturing, Explaining, and Demonstrating Lecturing, explaining, and demonstrating are common teacher activities in the direct-instruction approach (Mowbray & Perry, 2015). Effective teachers spend more time explaining and demonstrating new material than their less-effective counterparts do.

On some occasions we sit through boring lectures, yet on other occasions we have been captivated by a lecturer and learned a great deal from the presentation. Let's explore some guidelines for when lecturing is a good choice and some strategies for delivering an effective lecture. Here are some goals that lecturing can accomplish (Henson, 1988):

1. Presenting information and motivating students' interest in a subject
2. Introducing a topic before students read about it on their own, or giving instructions on how to perform a task
3. Summarizing or synthesizing information after a discussion or inquiry
4. Providing alternative points of view or clarifying issues in preparation for discussion
5. Explaining materials that students are having difficulty learning on their own

advance organizers Teaching activities and techniques that establish a framework and orient students to material before it is presented.

expository advance organizers Organizers that provide students with new knowledge that will orient them to the upcoming lesson.

comparative advance organizers Organizers that introduce new material by connecting it with the students' prior knowledge.

Questioning and Discussing It is necessary but challenging to integrate questions and discussion in teacher-centered instruction (Jiang, 2014; Robitaille & Maldonado, 2015). Teachers should respond to each student's learning needs while maintaining



CONNECTING WITH STUDENTS: Best Practices

Strategies for Lecturing

To ensure success when lecturing, follow these strategies:

1. *Be prepared.* Don't just "wing" a lecture. Spend time preparing and organizing what you will present.
2. *Make the lecture interesting and exciting by keeping it short and interspersing it with questions and activities.* For example, lecture for 10 or 15 minutes to provide the background information and framework for a topic, then place students in small discussion groups. Vary the pace of the lecture by interlacing it with related video clips, demonstrations, hand-outs, and/or activities for students. Think about what you can say that will motivate students' interest in a topic.
3. *Follow a designated sequence and include certain key components:*
 - Begin with advance organizers or previews of the topic.
 - Verbally and visually highlight any key concepts or new ideas (like the boldfaced key terms in this book). Use the blackboard, an overhead projector, or another large-display device.
 - Present new information in relation to what students already know about the topic.
 - Periodically elicit student responses to ensure that they understand the information up to that point and to encourage active learning.
 - At the end of the lecture, provide a summary or an overview of the main ideas.
 - Make connections to future lectures and activities.

the group's interest and attention. It also is important to distribute participation widely while also retaining the enthusiasm of eager volunteers. An additional challenge is allowing students to contribute while still maintaining the focus on the lesson. In a research review, asking a large number of questions and monitoring the responses of all students was one of the four specific practices linked to students' achievement (Coe et al., 2014).



CONNECTING WITH STUDENTS: Best Practices

Strategies for the Effective Use of Questions

Let's examine some effective strategies for using questions in the classroom:

1. *Use fact-based questions as entrées into thinking-based questions.* For example, in teaching a lesson on environmental pollution, the teacher might ask the fact-based question: "What are three types of environmental pollution?" Then she could follow with this thinking-based question: "What strategies can you think of for reducing one of these types of environmental pollution?" Don't over-use fact-based questions, because they tend to produce rote learning rather than learning for understanding.
2. *Avoid yes/no and leading questions.* Yes/no questions should be used only as a segue into more probing questions. For example, avoid questions like "Was environmental pollution responsible for the dead fish in the lake?" Keep these questions to a minimum, only occasionally using them as a warm-up for questions such as these: "How did the pollution kill the fish?" "Why do you think companies polluted the lake?" "What can be done to clean up environmental pollution?"
Also avoid asking leading questions such as "Don't you agree?" or other rhetorical questions such as "You do want to read more about environmental pollution, don't you?" These types of questions don't produce meaningful responses and simply hand the initiative back to the teacher.
3. *Leave enough time for students to think about answers.* Too often when teachers ask questions, they don't give students enough time to think. In one study, teachers waited less than one second, on average, before calling on a student to respond (Rowe, 1986)! In the same study, teachers waited only about one second, on average, for the student to respond before supplying the answer themselves. Such intrusions don't give students adequate time to construct answers. In the same study, teachers were subsequently

continued



CONNECTING WITH STUDENTS: Best Practices

Strategies for the Effective Use of Questions



What are some good teaching strategies for the effective use of questions?

instructed to wait three to five seconds to allow students to respond to questions. The increased wait time led to considerable improvements in responses, including better inferences about the materials and more student-initiated questions. Waiting three to five seconds or more for students to respond is not as easy as it might seem; it takes practice. But your students will benefit considerably from having time to think and construct responses.

4. *Ask clear, purposeful, brief, and sequenced questions.* Avoid being vague. Focus the questions on the lesson at hand. Plan ahead so that your questions are meaningfully tied to the topic. If your questions are long-winded, you

run the risk that they will not be understood, so briefer is better. Also plan questions so that they follow a logical sequence, integrating them with previously discussed material before moving to the next topic.

5. *Monitor how you respond to students' answers.* What should you do after a student responds to your question? Many teachers just say "Okay" or "Uh-huh" (Sadker & Zittleman, 2016). Instead, use the student's response as a basis for follow-up questions and engage the student or other students in a dialogue. Provide feedback that is tailored to the student's existing level of knowledge and understanding.
6. *Be aware of when it is best to pose a question to the entire class or to a particular student.* Asking the entire class a question makes all students in the class responsible for responding. Asking a specific student a question can make other students less likely to answer it. Some reasons to ask a question to a particular student are: (a) to draw an inattentive student into the lesson, (b) to ask a follow-up question of someone who has just responded, and (c) to call on someone who rarely responds when questions are asked to the class as a whole. Don't let a small group of assertive students dominate the responses; talk with them independently about continuing their positive responses without monopolizing class time. One strategy for giving students an equal chance to respond is to pull names from a cookie jar or check names off a class list as students respond (Weinstein & Romano, 2015).
7. *Encourage students to ask questions.* Praise them for good questions. Ask them "How?" and "Why?" and encourage them to ask "How?" and "Why?"

Mastery Learning Mastery learning involves learning one concept or topic thoroughly before moving on to a more difficult one. A successful mastery learning approach involves these procedures (Bloom, 1971; Carroll, 1963):

- Specify the learning task or lesson. Develop precise instructional objectives. Establish mastery standards (this typically is where *A* students perform).
- Break the course into learning units that are aligned with instructional objectives.
- Plan instructional procedures to include corrective feedback to students if they fail to master the material at an acceptable level, such as 90 percent correct. The corrective feedback might take place through supplemental materials, tutoring, or small-group instruction.
- Give an end-of-unit or end-of-course test that evaluates whether the student has mastered all of the material at an acceptable level.

Mastery learning gets mixed reviews. Some studies indicate that mastery learning is effective in increasing the time that students spend on learning tasks (Lalley & Gentile, 2009; Mitee & Obaitan, 2015), but others find less support for mastery learning (Bangert et al., 1983). Outcomes of mastery learning depend on the teacher's skill in planning and executing the strategy (Joyce et al., 2015).

mastery learning Involves learning one topic or concept thoroughly before moving on to a more difficult one.

One context in which mastery learning might be especially beneficial is remedial reading (Schunk, 2020). A well-organized mastery learning program for remedial reading allows students to progress at their own rates based on their skills, their motivation, and the time they have to learn.

Seatwork Seatwork refers to the practice of having all or a majority of students work independently at their seats. Teachers vary in how much they use seatwork as part of their instruction (Weinstein, 2015). Some teachers use it every day; others rarely use it. Figure 2 summarizes the challenges of seatwork for the teacher and the student. Learning centers are especially good alternatives to paper-and-pencil seatwork. Figure 3 provides some suggestions for learning centers. A computer station can be an excellent learning center.

Homework Another important instructional decision involves how much and what type of homework to give students (Fernandez-Alonso et al., 2015; Novak & Lynott, 2015; Valle et al., 2015). Comments by a leading researcher, Harris Cooper (2006, 2007, 2009), suggest that in the United States homework is the source of more friction between school and home than any other aspect of education. In the cross-cultural research discussed earlier that focused on Asian and American students, the time spent on homework was assessed (Chen & Stevenson, 1989). Asian students spent more time doing homework than American students did. For example, on weekends Japanese first-graders did an average of 66 minutes of homework, and American first-graders did only 18 minutes. Also, Asian students had a much more positive attitude about homework than American students did. And Asian parents were far more likely to help their children with their homework than American parents were. In 2018, Black teens and those from lower-income households were more likely to report being unable to complete homework because they lacked reliable access to a computer or Internet connection; it is likely that this “homework gap” was exacerbated by school closures during the COVID-19 pandemic (Auxier & Anderson, 2020).

In a research review, Cooper et al., (2006) concluded that overall research indicates there is a positive influence of homework on students’ achievement. The research review also indicated that the homework-achievement link is stronger in grades 7 to 12 than kindergarten to grade 6. In one study, Cooper (1998) collected data on 709 students in grades 2 through 4 and 6 through 12. In the lower grades, there was a significant negative relation between the amount of homework assigned and students’ attitudes, suggesting that elementary school children resent having to do homework. But in grades 6 and higher, the more homework students completed, the higher their achievement. It is not clear which was the cause and which the effect, though. Were really good students finishing more assignments because they were motivated and competent in academic subjects, or was completing homework assignments causing students to achieve more?

A key aspect of the debate about whether elementary school children should be assigned homework is the type of homework assigned. What is good homework? Especially for younger children, the emphasis should be on homework that fosters a love of learning and sharpens study skills. Short assignments that can be quickly completed

For the teacher

1. Keeping track of what the rest of the class is doing
2. Keeping students on-task
3. Dealing with the varying paces at which students work (“ragged” endings)
4. Selecting or creating seatwork that is clear and meaningful
5. Matching seatwork to students’ varying levels of achievement
6. Collecting, correcting, recording, and returning seatwork assignments

For the student

1. Completing assigned work on their own
2. Understanding how and when to obtain the teacher’s help
3. Understanding the norms for assisting peers
4. Learning how to be effective in obtaining help from peers

FIGURE 2 CHALLENGES OF SEATWORK FOR TEACHERS AND STUDENTS



DIVERSITY



RESEARCH



DEVELOPMENT

Science	Social studies	Mathematics	Art	Writing	Computer
Simple experiments with lab sheets Observations over time with recording forms Exploring properties of objects and classifying them	Recreating items used by different civilizations Creating charts or graphs of population trends Map making	Math “challenges” and puzzles Manipulative activities	Holiday or thematic projects Crafts related to curriculum studies (quilting, quilling, origami, etc.)	Class story writing (e.g., add-on stories) Rewrites of literature Writing plays or puppet shows	Content-related programs Simulations Story writing

FIGURE 3 SUGGESTIONS FOR LEARNING CENTERS

should be the goal. With young children, long assignments that go uncompleted or completed assignments that bring a great deal of stress, tears, and tantrums should be avoided. Too often teachers assign homework that duplicates without reinforcing material that is covered in class. Homework should be an opportunity for students to engage in creative, exploratory activities, such as doing an oral history of one's family or determining the ecological effects of neighborhood business. Instead of memorizing names, dates, and battles of the Civil War as a homework assignment, students might write fictional letters from Northerners to Southerners, expressing their feelings about the issues dividing the nation. The homework assignments should be linked to the next day's class activities to emphasize to students that homework has meaning and is not just a plot to make them miserable. Homework also should have a focus. Don't ask students to write an open-ended theme from a novel the class is reading. Rather, ask them to select a character and explain why he or she behaved in a particular way.



RESEARCH

In Cooper et al., (2006) review of research, homework began to have a payoff in middle school. How can homework have little or no effect in elementary school yet be so beneficial in middle and high school? In the higher grades, it is easier to assign focused, substantive homework that requires students to integrate and apply knowledge—the type of homework that promotes learning (Corno, 1998). Also, by high school, students have resigned themselves to the routine of homework. Working hard after school and having good study skills are more accepted by middle and high school students.

Some educational psychologists argue that the main reason homework has not been effective in elementary school is that it has focused too much on subject matter and not enough on developing attitudes toward school, persistence, and responsible completion of assignments (Corno, 1998). They stress that it is not homework per se that benefits students but, rather, the opportunities it provides for the student to take responsibility. They think that teachers need to inform parents about guiding their children in these aspects of doing their homework: setting goals, managing their time, controlling their emotions, and checking their work, rather than playing avoidance games or leaving hard work for last. Teachers and parents can use homework in the early grades to help children wrestle with goal setting and follow-through.

Cooper and his colleagues (Cooper, 2009; Cooper & Patall, 2007; Cooper et al., 2006) also have found the following:

- Homework has more positive effects when it is distributed over a period of time rather than done all at once. For example, doing 10 math problems each night for five nights is better than doing 50 over the weekend.
- Homework effects are greater for math, reading, and English than for science and social studies.
- For middle school students, one or two hours of homework a night is optimal. High school students benefit from even more hours of homework, but it is unclear what a maximum number of hours ought to be.

Homework can be a valuable tool for increasing learning, especially in middle and high school (Cooper, 2006, 2007, 2009). However, it is important to make homework meaningful, monitor it, and give students feedback about it.



RESEARCH

In the last decade, there has been increasing research interest in parents' roles in students' homework (Madjar et al., 2016; Moroni et al., 2015; Nunez et al., 2015; Silinskas et al., 2015). A research review concluded that positive achievement outcomes occur only when parents are directly involved in assisting their children during homework tasks (Aries & Cabus, 2015). In this review, positive outcomes especially occurred when the parental assistance involved engaging in meta-strategies or support for children's understanding of the homework. Also, in this review, when parents simply facilitated, structured, or emotionally supported their children's homework, achievement outcomes were either indecisive or negative. Communication between teachers and families is key for helping students who are not completing their homework, which became an even greater concern during the pandemic (Clausen et al., 2020).

EVALUATING TEACHER-CENTERED INSTRUCTION

Research on teacher-centered instruction has contributed many valuable suggestions for teaching, including these:

- Be an organized planner, create instructional objectives, and spend initial time orienting students to a lesson.
- Have high expectations for students' progress and ensure that students have adequate academic learning time.
- Use lecturing, explaining, and demonstrating to benefit certain aspects of students' learning.
- Engage students in learning by developing good question-asking skills and getting them involved in class discussion.
- Have students do meaningful seatwork or alternative work to allow individualized instruction with a particular student or a small group.
- Give students meaningful homework to increase their academic learning time and involve parents in students' learning.

Advocates of the teacher-centered approach especially believe that it is the best strategy for teaching basic skills, which involve clearly structured knowledge and skills (such as those needed in English, reading, math, and science). Thus, in teaching basic skills, the teacher-centered approach might consist of a teacher explicitly or directly teaching grammar rules, reading vocabulary, math computations, and science facts.

Teacher-centered instruction has been criticized for several reasons, however. Critics say that teacher-centered instruction often leads to passive, rote learning and inadequate opportunities to construct knowledge and understanding (McCall, 2007; Serin, 2018). They also criticize teacher-centered instruction as producing overly structured and rigid classrooms, inadequate attention to students' socioemotional development, external rather than internal motivation to learn, too much reliance on paper-and-pencil tasks, few opportunities for real-world learning, and too little collaborative learning in small groups. Such criticisms often are leveled by advocates of learner-centered planning and instruction, a topic we will explore next.

Review, Reflect, and Practice

2 Identify important forms of teacher-centered instruction.

REVIEW

- What are some useful tools in teacher-centered lesson planning?
- What is direct instruction?
- What are some good teacher-centered instructional strategies?
- What are some pros and cons of teacher-centered instruction?

REFLECT

- As a student, have you ever wished that a teacher used more (or less) teacher-centered instruction? What lessons can you draw from this for your own work as a teacher?

PRAXIS™ PRACTICE

1. Mr. McGregor has assigned his students to write an essay explaining the impact of the use of the atomic bomb during World War II. Which cognitive level of Bloom's taxonomy is best illustrated by this assignment?
 - a. analysis
 - b. application
 - c. comprehension
 - d. knowledge

continued

Review, Reflect, and Practice

2. A first-grade teacher spends about a quarter of her instructional time on mathematics and assigns at least an hour of math homework each weekend. This math class is most typical of instructional methods in
 - a. the United States
 - b. France
 - c. Japan
 - d. Germany
3. Ms. Davidson is giving a lecture to introduce her students to a unit on the American Revolution. She begins with a little-known fact to pique her students' interest. She then distributes a brief outline of what her lesson will cover. She lectures from carefully prepared notes, using visual aids to help her students understand the material and keep their interest. Periodically during the lecture she asks questions to make sure her students are attending to, and understanding, the material. She calls on the first person to raise her hand to answer the question. If this person does not answer immediately, she supplies the answer herself so that the students do not become confused and the lesson does not drag on. What should Ms. Davidson do differently?
 - a. Allow her students more time to answer questions.
 - b. Avoid asking questions during the lecture.
 - c. Be more spontaneous in her class presentations.
 - d. Avoid using distracting visual aids.
4. Ms. Bancroft likes to lecture most of the time, makes sure that her students stay on task when they are in the classroom, and has students do meaningful seatwork. Ms. Bancroft is following which of these approaches?
 - a. cognitive constructivist
 - b. social constructivist
 - c. direct instruction
 - d. individualized

Please see answer key at end of chapter

LG 3 Discuss important forms of learner-centered instruction.

3 LEARNER-CENTERED LESSON PLANNING AND INSTRUCTION

Learner-Centered Principles

Some Learner-Centered Instructional Strategies

Evaluating Learner-Centered Strategies

Just as the behavioral approaches provide the conceptual underpinnings for teacher-centered lesson planning and instruction, the information processing and constructivist approaches form the theoretical backdrop for learner-centered lesson planning and instruction. In this section, we will explore the principles and strategies used in learner-centered instruction.

LEARNER-CENTERED PRINCIPLES

Learner-centered lesson planning and instruction move the focus away from the teacher and toward the student (McCombs, 2010, 2015). In one large-scale study, students' perceptions of a positive learning environment and interpersonal relationships with the teacher—factors associated with learner-centered instruction—were important in enhancing students' motivation and achievement (McCombs, 2001).

Increased interest in learner-centered principles of lesson planning and instruction has resulted in a set of guidelines called *Learner-Centered Psychological Principles*:

A Framework for School Reform and Redesign (Presidential Task Force on Psychology and Education, 1992; Work Group of the American Psychological Association Board of Educational Affairs, 1997). The guidelines were constructed and are periodically revised by a prestigious group of scientists and educators from a wide range of disciplines and interests. These principles have important implications for the way teachers plan and instruct, as they are based on research on the most effective ways children learn.

The Work Group of the American Psychological Association Board of Educational Affairs (1997) stresses that research in psychology relevant to education has been especially informative, producing advances in our understanding of cognitive, motivational, and contextual aspects of learning. The work group states that the learner-centered psychological principles it has proposed are widely supported and are being increasingly adopted in many classrooms. The principles emphasize the active, reflective nature of learning and learners. According to the work group, education will benefit when the primary focus is on the learner.

The fourteen learner-centered principles can be classified in terms of four main sets of factors: cognitive and metacognitive, motivational and instructional, developmental and social, and individual differences (Work Group of the American Psychological Association Board of Educational Affairs, 1997). Figure 4 describes the 14 learner-centered principles.



John Mahoney, teaching math to one of his students. Another student of Mahoney's, Nicole Williams, says, "He's not going to tell you the answer. He's going to make you think" (Wong Briggs, 2004, p. 6D). *What are some important cognitive and metacognitive factors in learner-centered instruction?*

John F. Mahoney

SOME LEARNER-CENTERED INSTRUCTIONAL STRATEGIES

In other chapters we discussed a number of strategies that teachers can consider in developing learner-centered lesson plans. These especially include the teaching strategies based on the theories of Piaget and Vygotsky, constructivist and social constructivist aspects of thinking, and learning and cognition in the content areas. To provide you with a further sense of learner-centered strategies that you can incorporate into your lesson planning, we will examine problem-based learning, essential questions, and discovery learning.

Problem-Based Learning Problem-based learning emphasizes real-life problem solving (Ali, 2019; Barber et al., 2015; Winfrey Avant & Bracy, 2015). A problem-based curriculum exposes students to authentic problems like those that crop up in everyday life. Problem-based learning is a learner-centered approach that focuses on a problem to be solved through small-group efforts. Students identify problems or issues that they wish to explore, then locate materials and resources they need to address the issues or solve the problems. Teachers act as guides, helping students to monitor their own problem-solving efforts.

An example of a problem-based learning project involves sixth-grade students in exploring an authentic health problem in their local community: the causes, incidence, and treatment of asthma and its related



What are some learner-centered psychological principles?

Photodisc/SuperStock

Cognitive and Metacognitive Factors: These principles highlight learners' intellectual capacities and how the learning process is facilitated by contexts that support thinking strategies.

1. **Nature of the learning process**
The learning of complex subject matter is most effective when it is an intentional process of constructing meaning from information and experience.
2. **Goals of the learning process**
The successful learner, over time and with support and instructional guidance, can create meaningful, coherent representations of knowledge.
3. **Construction of knowledge**
The successful learner can link new information with existing knowledge in meaningful ways.
4. **Strategic thinking**
The successful learner can create a repertoire of thinking and reasoning strategies to achieve complex goals.
5. **Thinking about thinking**
Higher-order strategies for selecting and monitoring mental operations facilitate creative and critical thinking.
6. **Context of learning**
Learning is influenced by environmental factors, including culture, technology, and instructional practices.

Motivational and Instructional Factors: These principles highlight the roles of motivation and emotion in learning, focusing on how contexts can meet individual learners' needs for control and choice.

7. **Motivational and emotional influences on learning**
What and how much is learned is influenced by the learner's motivation. Motivation to learn, in turn, is influenced by the learner's emotional states, beliefs, interests, goals, and habits of thinking.
8. **Intrinsic motivation to learn**
The learner's creativity, higher-order thinking, and natural curiosity all contribute to motivation to learn. Intrinsic motivation is stimulated by tasks of optimal novelty and difficulty, relevant to personal interests and providing for personal choice and control.
9. **Effects of motivation on effort**
Acquisition of complex knowledge and skills requires extended learner effort and guided practice. Without learners' motivation to learn, the willingness to exert this effort is unlikely without coercion.

Developmental and Social Factors: These principles focus on how learner development and interpersonal interactions can influence learning.

10. **Developmental influences on learning**
As individuals develop, there are different opportunities and constraints for learning. Learning is most effective when development within and across physical, cognitive, and socioemotional domains is taken into account.
11. **Social influences on learning**
Learning is influenced by social interactions, interpersonal relations, and communication with others.

Individual Difference Factors: These principles emphasize the importance of setting high standards and using assessment to adapt instruction to meet the needs of diverse learners.

12. **Individual differences in learning**
Learners have different strategies, approaches, and capabilities for learning that are a function of prior experience and heredity.
13. **Learning and diversity**
Learning is most effective when differences in learners' linguistic, cultural, and social backgrounds are taken into account.
14. **Standards and assessment**
Setting appropriately high and challenging standards and assessing the learner as well as learning progress—including diagnostic, process, and outcome assessment—are integral parts of the learning process.

FIGURE 4 LEARNER-CENTERED PSYCHOLOGICAL PRINCIPLES

conditions (Jones et al., 1997). Students learn how environmental conditions affect their health and share this understanding with others. The project integrates information from many subject areas, including health, science, math, and the social sciences. One study found that problem-based learning was positive associated with high school students' critical thinking skills in math courses (Widyatiningtyas et al., 2015). Moreover, a recent study found that integrating augmented reality into problem-based learning activities increased seventh-grade students' learning achievement above and beyond problem-based learning activities' benefits compared to traditional teaching; integrating augmented reality also promoted greater positive attitudes toward physics (Fidan & Tuncel, 2019).

In a meta-analysis that compared problem-based learning with traditional teaching methods, problem-based learning had a more positive effect on students' attitudes (Batdi, 2014). In one study, 20 eleventh-grade chemistry students were randomly assigned to either a teacher-centered or problem-based learning experience to learn about the effects of temperature, concentration, and pressure on cell potential (Tarhan & Acar, 2007). Interviews with the students following the instruction revealed that the students in both conditions were equally successful in gaining an understanding of the topic, but that students in the problem-based learning condition were more motivated, self-confident, and interested in solving problems.

Essential Questions Essential questions are questions that reflect the heart of the curriculum, the most important things that students should explore and learn. For example, in one lesson the initial essential question was “What flies?” Students explored the question by examining everything from birds, bees, fish, and space shuttles to the notion that time flies and ideas fly. The initial question was followed by other questions, such as “How and why do things fly in nature?” “How does flight affect humans?” and “What is the future of flight?”

Essential questions like these perplex students, cause them to think, and motivate their curiosity (Stronge, 2018; Wiggins & Wilbur, 2015; Wilhelm, 2014). Essential questions are creative choices. With just a slight change, a lackluster question such as “What was the effect of the Civil War?” can become the thought-provoking question “Is the Civil War still going on?”

Discovery Learning Discovery learning is learning in which students construct an understanding on their own. Discovery learning stands in contrast to the direct-instruction approach discussed earlier, in which the teacher directly explains information to students. In discovery learning, students have to figure out things for themselves. Discovery learning meshes with the ideas of Piaget, who once commented that every time you teach a child something you keep the child from learning.

As teachers began to use discovery learning, they soon found that for it to be effective as a systematic instruction approach it needed to be modified. This led to the development of **guided discovery learning**, in which students are still encouraged to construct their understanding, but with the assistance of teacher-guided questions and directions (Janssen et al., 2014).

A research review indicated that guided discovery learning was superior to pure discovery learning in every case (Mayer, 2004a). In this review, it also was concluded that constructivist learning is better supported by curricular focus than by pure discovery.

EVALUATING LEARNER-CENTERED STRATEGIES

The learner-centered approach to lesson planning and instruction is positive in many ways (McCombs, 2015). The fourteen learner-centered principles developed by the American Psychological Association task



RESEARCH

essential questions Questions that reflect the heart of the curriculum, the most important things that students should explore and learn.

discovery learning Learning in which students construct an understanding on their own.

guided discovery learning Learning in which students are encouraged to construct their understanding with the assistance of teacher-guided questions and directions.

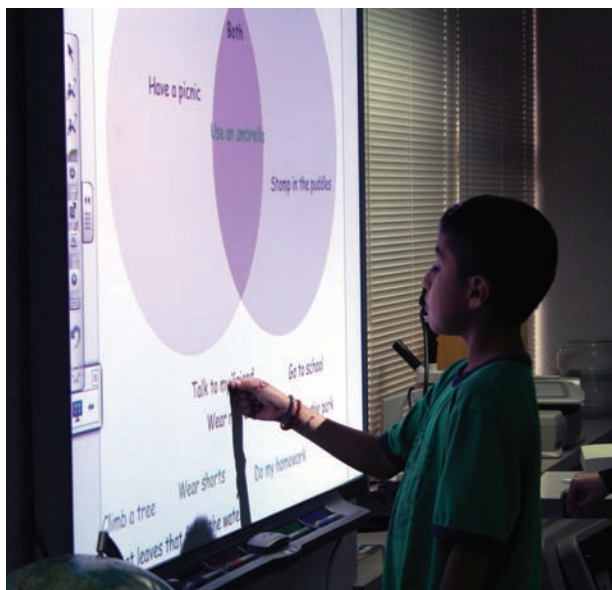


What is guided discovery learning?

Steve Debenport/Getty Images

force can be very helpful in guiding students' learning. The principles encourage teachers to help students actively construct their understanding, set goals and plan, think deeply and creatively, monitor their learning, solve real-world problems, develop more positive self-esteem and control their emotions, be internally motivated, learn in a developmentally appropriate way, collaborate effectively with others (including diverse others), evaluate their learner preferences, and meet challenging standards.

Critics of learner-centered instruction argue that it gives too much attention to the process of learning (such as learning creatively and collaboratively, like through discovery learning) and not enough to academic content (such as the facts of history, like through lecture) (Hirsch, 1996). Some critics stress that learner-centered instruction works better in some subjects than in others (Feng, 1996). They say that in areas with many ill-defined problems, such as the social sciences and humanities, learner-centered instruction can be effective. However, they believe that in well-structured knowledge domains such as math and science, a teacher-centered structure works better. Critics also say that learner-centered instruction is less effective at the beginning level of instruction in a field because students do not have the knowledge to make decisions about what and how they should learn. And critics stress that there is a gap between the theoretical level of student-centered learning and its actual application (Airasian & Walsh, 1997). The consequences of implementing learner-centered strategies in the classroom are often more challenging than anticipated.



Students using interactive whiteboards at the Research Center for Educational Technology's AT&T classroom at Kent State University.
Research Center for Educational Technology, Kent State University

In a study, the Minnesota New Country School (MNCS), a learner-centered school, was evaluated (Aslan & Reigeluth, 2015/2016). This school uses a self-directed, project-based learning approach with each student choosing an advisor until they graduate. The advisors and administrators in the school all said that the number one challenge in the school was changing students' mindsets from being directed by teachers to engaging in self-direction.

Although we have presented teacher-centered and learner-centered planning and instruction in separate sections, don't think of them as always being either/or approaches. Many effective teachers use some of both in making the classroom a positive learning environment for children (Yeh, 2009). Research on the choice and sequencing of learning activities in the classroom indicates that using both constructivist and direct instruction approaches is often more effective than using either approach alone (Darling-Hammond et al., 2005).

Consider high school history teacher Robert Bain's (2005) description of how students initially spent time in learner-centered instruction working on constructing accounts of Columbus' voyage to America and struggled with the question of how the anniversaries of the voyage were celebrated. Bain then used teacher-centered direct instruction by lecturing about leading historians' current thinking about the topic. Also, a study of a ninth-grade technology-enhanced classroom revealed that both student-centered and teacher-centered instruction promoted students' conceptual understanding and provided students with different opportunities to engage in learning (Wu & Huang, 2007).

Interactive whiteboards, a relatively new instructional technology, can be used in both teacher-centered and student-centered instruction (Swan et al., 2009). Interactive whiteboards are being used in many schools to replace the traditional blackboard and chalk. This new technology allows teachers and students to interact with content projected from a computer screen onto a whiteboard surface. Virtually anything that can be done on a computer can be done on an interactive whiteboard, with the advantage that drawing, marking, and highlighting of computer-based output is supported. A whole class can follow interactions, and lessons can be saved and replayed.

Some teachers were asked to describe their most effective learner-center strategies. Following are their responses.

EARLY CHILDHOOD Learner-centered strategies are common in my preschool classroom. For example, each morning children are given time for free play in which they play dress-up, pretend to cook in a mock kitchen, and build structures using Legos. Through these activities, children not only learn important skills, but they are also instilled with a love of school.



—MISSY DANGLER, *Suburban Hills School*

ELEMENTARY SCHOOL: GRADES K-5 I use guided discovery learning in almost all of my science classes. The children work in groups, are given guidance at the beginning of the lesson and then assigned a task to do in order to find the answer to a question. When students ask questions, I refer them to an instruction sheet I have given them or ask questions to guide their thinking. If a student asks, “Can we do this?” I reply, “Try it and see!” I also encourage students to count on other members of their group to uncover answers to questions and use phrases such as “Many heads are better than one.”



—JANINE GUIDA POUTRE, *Clinton Elementary School*

MIDDLE SCHOOL: GRADES 6-8 In order to master new vocabulary, I often have my students quiz each other on vocabulary words. It is a noisy activity, but the students love it and learn at the same time.



—MARGARET REARDON, *Pocantico Hills School*

HIGH SCHOOL: GRADES 9-12 I prefer being the “guide on the side,” not the “sage on the stage,” when it comes to teaching. I did a writing workshop in which I presented a short mini-lesson each day, and my students chose how to spend the rest of the period. The students set their own goals, and their grade was partly based on achievement of those goals. I conferenced one-on-one with each student daily to monitor how they were doing; students loved the freedom.



—JENNIFER HEITER, *Bremen High School*



CONNECTING WITH STUDENTS: Best Practices Strategies for Using Learner-Centered Instruction

Following are some effective ways to incorporate learner-centered instruction into your teaching:

1. *Become familiar with the learner-centered psychological principles and incorporate them in your lesson planning and teaching.*
2. *Focus on the whole child.* Pay attention to motivational and affective factors, and developmental and social factors, in addition to cognitive factors. In *Through the Eyes of Teachers*, you can read about elementary-school

science teacher Luis Recalde and how he focuses on the whole child and makes learning fascinating for students.

THROUGH THE EYES OF TEACHERS Fostering Learning, Unity, and Civic Pride

Luis Recalde, a fourth- and fifth-grade science teacher at Vincent E. Mauro Elementary School in New Haven, Connecticut, uses every opportunity to make science fascinating and motivating

continued



CONNECTING WITH STUDENTS: Best Practices

Strategies for Using Learner-Centered Instruction

for students to learn. Recalde infuses hands-on science experiences with energy and enthusiasm. To help students get a better sense of what it is like to be a scientist, he brings lab coats to the classroom for students to wear. He holds science fair workshops for teachers and often gives up his vacation time to help students with science projects. He started soccer teams and gardens to foster unity and civic pride among African American and Latinx students. An immigrant himself, he knows the importance of fostering positive relations among students from different ethnic groups.

3. *Use problem-based learning, essential questions, and guided discovery learning in your teaching.*
4. *Remember that the most effective teachers don't use direct instruction or learner-centered instruction exclusively but instead use both to make the classroom a positive learning environment for students.*



Elementary school science teacher Luis Recalde holds up a seaweed specimen in one of the hands-on, high-interest learning contexts he creates for students.

Stan Godlewski

Review, Reflect, and Practice

3 Discuss important forms of learner-centered instruction.

REVIEW

- What is learner-centered lesson planning? Summarize the APA's fourteen learner-centered principles.
- How do problem-based learning, essential questions, and discovery learning each embody learner-centered principles?
- What are some pros and cons of learner-centered instruction?

REFLECT

- As a student, have you ever wished that a teacher would use more (or less) learner-centered instruction? What lessons can you draw from this for your own work as a teacher?

PRAXIS™ PRACTICE

1. Joan just received a *D* on her science exam. "I knew it," she states, "I've never been any good at science." Which set of factors of the APA's learner-centered principles is best exemplified by Joan's statement?
 - a. cognitive and metacognitive
 - b. developmental and social
 - c. individual differences
 - d. motivational and emotional
2. Mr. Williams wants his third-grade students to understand the purpose of blubber in marine mammals. He sets up an experiment for his students using ice water, latex gloves, and lard. First, students put their gloved hands in ice water until they cannot keep them

Review, Reflect, and Practice

there any longer. Other students time how long they keep their hands submerged. Then, students put their gloved hands in bags of lard and submerge their hands again. Again, other students time how long they keep their hands submerged. All of the students are able to keep their hands in the ice water longer with their hands in the lard than they can when their hands are merely gloved. What learner-centered instructional strategy has Mr. Williams used?

- discovery learning
 - essential questions
 - guided discovery
 - problem-based learning
3. Ms. Flanagan, who takes a teacher-centered approach, just observed Mr. Houston's constructivist teaching in his seventh-grade math class. Ms. Flanagan is likely to be most critical of which of the following she observed?
- Mr. Houston assigning students homework
 - Mr. Houston encouraging students to construct their own math problem-solving strategies
 - Mr. Houston having high expectations for student learning
 - Mr. Houston using technology to help students learn

Please see answer key at end of chapter

4 TECHNOLOGY AND EDUCATION

The Technology Revolution
and the Internet

Standards for Technology-
Literate Students

Teaching, Learning, and
Technology

So far in this chapter we have described many aspects of planning and instruction. In contemporary society, technology plays important roles in planning and instruction. Three important ways that technology affects curriculum planning are: (1) as a learning goal for students to develop certain technology competencies; (2) as a resource for curriculum planning through the extensive materials that are available on the Internet; and (3) as tools that improve students' ability to learn through techniques such as simulation and visualization in science and text analysis in literature, as well as software that encourages reflection and provides models of good performance (Darling-Hammond et al., 2005).

Technology is such an important theme in education that it is woven throughout this book. In each chapter, you come across one or more discussions of technology related to the chapter's contents. In other chapters you have studied topics such as technology and children's vocabulary development, technology as a means of connecting with students around the world, computer-supported intentional learning environments and content-specific applications of technology in reading, writing, math, science, and social studies. Here we will explore the technology revolution and the Internet, standards for technology-literate students, and teaching and learning with technology.

THE TECHNOLOGY REVOLUTION AND THE INTERNET

Students today are growing up in a world that is far different technologically from the world in which their parents and grandparents were students. If students are to be adequately prepared for tomorrow's jobs, technology must be an integral part of schools and classrooms (Maloy et al., 2017; Roblyer, 2016).

LG 4 Summarize how to effectively use technology to help children learn.



TECHNOLOGY

Thinking Back/Thinking Forward

Technology and children's vocabulary development. Connect to "Cognitive and Language Development."

Technology connections with students around the world. Connect to "Sociocultural Diversity."

Computer-supported intentional learning environments. Connect to "Social Constructivist Approaches."

Content-specific applications of technology in reading, writing, math, science, and social studies. Connect to "Learning and Cognition in the Content Areas."



A student creating a multimedia program in the Research Center for Educational Technology's AT&T classroom at Kent State University.

Research Center for Educational Technology, Kent State University

Thinking Back/Thinking Forward

Collaborative learning is reflected in teaching strategies such as the use of cooperative learning and small-group work. Connect to "Social Constructivist Approaches."

Internet The core of computer-mediated communication; a system of computer networks that operates worldwide.

cloud computing Delivery of services such as servers, storage, and applications to an organization's computers and devices through the Internet.

Web A system for browsing Internet sites that refers to the World Wide Web; named the Web because it is composed of many sites that are linked together.

The Internet The **Internet** is a system of computer networks that operates worldwide. As the core of computer-mediated communication, the Internet is playing an important role in the technology revolution, especially in schools. In many cases, the Internet has more current, up-to-date information than textbooks. Nearly 100 percent of public schools in the United States are now connected to the Internet, and schools are increasingly using **cloud computing**, where services such as servers, storage, and applications are delivered to an organization's computers and devices through the Internet.

The Internet did not become the common portal it is today until the introduction of the World Wide Web (the **Web**) in 1991. The **Web** is a system for browsing Internet sites. It is named the Web because it is made of many sites linked together. The Web presents the user with documents, called Web pages, full of links to other documents or information systems. Selecting one of these links, the user can access more information about a particular topic. Web pages include text as well as multimedia (images, video, animation, and sound—all of which can be accessed by students with a click or tap for interactions). Web indexes and search engines such as Google can help students find the information they are seeking by examining and collating a variety of sources.

The Internet can be a valuable tool for helping students learn. However, it has some potential drawbacks. To use it effectively with your students, you will have to know how to use it and feel comfortable with it, as well as have up-to-date equipment and software. In addition, concerns have been raised about students accessing pornographic material and biased Web sites, as well as the questionable accuracy of some information gleaned from the Internet. Many of these problems are solved by installing firewalls or blocking software on school servers.

When used effectively, however, the Internet expands access to a world of knowledge and people that students cannot experience in any other way. Here are some effective ways that the Internet can be used in classrooms:

- *Navigating and integrating knowledge.* Teachers can focus on authentic data for problem solving and inquiry, such as temperature records or census data. Also, it can be helpful to reconcile information from multiple sources as well as teaching and modeling evaluation of information, analysis of source credibility, and documentation of Internet resources.
- *Collaborative learning.* One of the most effective ways to use the Internet in your classroom is through project-centered activities. Blog publishing (allowing moderated comments) can be a good collaborative online activity. Another collaborative use of the Internet is to have a group of students conduct a survey on a topic, put it on the Internet, and hope to obtain responses from many parts of the world. They can organize, analyze, and summarize the data they receive, and then share them with other classes around the world. Yet another type of collaborative learning project involves sending groups of students on Internet "scavenger hunts" to find information and/or solve a problem. Online citizen science projects also are inherently collaborative with data being pooled online for analysis. In the classroom, Web publishing and online videos can be highly collaborative activities with activities with different groups assigned to different roles and responsibilities (e.g., art department, script writers, technical team).
- Kent Innovation High School in Michigan extensively uses project-based learning and the Internet (Langel, 2015). Every student has a laptop and the entire school is invested in the use of Google Drive, where students can store designs, drawings, recordings, stories, videos, and so on. Teachers use Google Drive to introduce, trace, and plan projects with students, who use it for generating ideas, planning, and orchestrating their own projects.
- *Computer-mediated communications (CMC).* An increasing number of innovative educational projects include the use of computer-mediated communications. For example, there are many online sites (such as www.studentsoftheworld.info)

and www.tesol.net/teslpnpl.html) set up to allow teachers and students to correspond with “pen pals” around the world. Some of these even provide safe access to more innovative CMC forms such as chats and blogs.

- *Improving teachers’ knowledge and understanding.* Two excellent Internet resources for teachers are the Educational Resources Information Center (ERIC at <https://eric.ed.gov/>) and the Educators’ Reference Desk (www.eduref.org), which provide free information about a wide range of educational topics. The ERIC database gives abstracts of more than 1 million journal and nonjournal educational papers going back to 1966 and the full text of over 100,000 papers from educational conferences. The Educators’ Reference Desk gives easy access to over 2,000 lesson plans and over 3,000 links to online education information. Other excellent Internet resources for teachers include Harvard’s Websites for Educators (www.gse.harvard.edu/library/educator-resources), a selective list of topically organized websites that support the practice of educators, and the PBS TeacherLine (<http://teacherline.pbs.org/teacherline>), which provides links to teaching resources and online professional development courses for teachers.
- NASA for educators (<https://www.nasa.gov/stem/foreducators/k-12/index.html>) has an extensive set of resources geared to teachers at all levels and in all subject areas as well as professional development materials to help teachers use these resources effectively.

Increasing numbers of teachers are using social media as an important tool for professional networking and development. Many Twitter chats are devoted to teachers in certain areas, grade levels, or content areas. Teachers often report that this activity is a useful and meaningful source of professional support and ideas. A list of educator Twitter chats can be found at www.iste.org/explore/article/Detail?articleid=7.

Graphics and Presentation Graphics software and PowerPoint can improve many presentations by enhancing the content and attractiveness of the visual aids you use to teach students. Software graphics packages can be used to copy and create images, formulate charts and graphs, set the timing of material, and compose effective visual presentations. When used competently, PowerPoint and other graphics software packages can improve the organization of content and make it more dynamic.



A high school student creating a PowerPoint presentation in a math class.
Herwig Vergult/AFP/Getty Images

STANDARDS FOR TECHNOLOGY-LITERATE STUDENTS

If students are to be adequately prepared for tomorrow’s jobs, schools must take an active role in ensuring students become technologically literate. Most national standards recognize this. For example, the National Council of Teachers of English/International Reading Association Standards for the English Language Arts (NCTE/IRA, 2012) include the following: “Students use a variety of technological and informational resources (e.g., libraries, databases, computer networks, video) to gather and synthesize information and to create and communicate knowledge,” and a major theme in the National Council for the Social Studies, Curriculum Standards for the Social Studies (NCSS, 2010) is “Science, Technology, and Society.”

The International Society for Technology in Education (ISTE) has developed technology standards for students (ISTE, 2016) and



A student creating a bar graph.
Michael Scott/McGraw-Hill Education

educators (ISTE, 2017). ISTE's (2016) technology standards for students involve being a(an):

- *Empowered Learner*. Students actively use technology to reach learning goals.
- *Digital Citizen*. Students demonstrate responsibility and are ethical in their use of technology.
- *Knowledge Constructor*. Students use a variety of resources and digital tools to construct knowledge, become more creative, and engage in meaningful learning.
- *Innovative Designer*. Students use various technologies to solve problems and craft useful and imaginative solutions to these problems.
- *Computational Thinker*. Students develop strategies in using technology to create solutions and test them.
- *Creative Communicator*. Students communicate effectively and think creatively in their use of digital tools to attain goals.
- *Global Collaborator*. Students use technology to widen their perspectives and enhance their learning by connecting with others locally and globally.

In addition, ISTE provides performance indicators for achieving these standards at four levels: prekindergarten through grade 2, grades 3 through 5, grades 6 through 8, and grades 9 through 12. ISTE also includes examples and scenarios to illustrate how technology literacy can be integrated across the curriculum at each of these levels.



At the second level (grades 3 through 5), a teacher might make extensive use of Internet resources. She could use the Global Learning and Observations for a Better Environment (GLOBE) Web site (www.globe.gov) to engage students in making environmental observations around the school, reporting the data to a processing facility through GLOBE, and using global images created from their data to examine local environmental issues.

Population growth and urban planning are the focus of a social studies technology-based learning activity in grades 9 through 12. The activity challenges students to find sources online and elsewhere that describe real-world population dilemmas. The activity can be altered to address different cities and regions worldwide. In small groups in class, students can discuss problems that may occur as a result of a city being heavily populated. They can be asked to project what problems a city such as Tokyo is likely to face in terms of population growth in the year 2050.



Students at Christopher Maddox High School in Park City, Utah, present a daily, live TV broadcast to more than 1,200 peers and 80 adults. The students create a program that highlights school news and events and have full responsibility for writing scripts, operating cameras, directing segments, and handling other aspects of producing a TV broadcast.

UrbanCowE+/Getty Images

TEACHING, LEARNING, AND TECHNOLOGY

A special concern is how technology can be used to improve teaching and learning (Maloy et al., 2017; Roblyer, 2016). Following are some effective strategies for using technology in the classroom.

Using Technology to Improve Students' Understanding In the last several decades at the Educational Technology Center at Harvard University, a number of educators have focused on developing ways to use technology to improve students' understanding. Martha Stone Wiske has especially been instrumental in creating ways to incorporate technology into classroom contexts that transform student learning. Wiske et al., (2005) described how to more effectively use technology to teach for understanding by considering (1) the topics that are worth

understanding, (2) what students should understand about such topics, (3) how students develop and demonstrate understanding, (4) how students and teachers assess understanding, and (5) how students and teachers learn together. These five aspects of understanding are based on ideas developed at Harvard by David Perkins, Howard Gardner, and Vito Perrone. Following are Wiske et al., (2005) views of how to use technology to enhance understanding:

1. *Evaluate which topics are worth understanding.* Technology is especially appropriate for generating worthwhile and interesting learning topics. The Internet provides a wealth of information about virtually every topic imaginable that can be mined to generate new topics or expand what students are studying as part of the curriculum. The wide range of information provided by the Internet allows students to learn more about their own interests and ideas and carve a unique pathway in learning about a topic, instead of following cookie-cutter steps in a traditional textbook or workbook (Roblyer, 2016). Wiske et al., (2005) also point out that technology can often be used effectively to teach where problem spots emerge every year: “examples include heat and temperature or weight and density in science, ratios in mathematics, and stereotypes in history and social studies classes” (p. 28). These topics, which many students struggle to understand, are central to the subject matter and are frequently more easily understood through the use of technology.
2. *Think about what students should understand about a topic.* When teachers consider using technology in the classroom, it is important for them to think about the learning goals they have for their students. Goals might include learning a new concept or applying a key concept to relevant situations. A goal related to technology might be to understand how to find and critically examine information on the Internet that is relevant to a classroom topic. Generating goals in this manner reminds the teacher that “surfing the Web” is “not an end in itself” but rather a way to use “technology to accomplish meaningful work” (Wiske et al., 2005, p. 44).
3. *Pay attention to how students develop and demonstrate understanding.* Use technology to help students “stretch their minds” and understand something in ways that they never did before. In improving students’ understanding, Wiske et al., (2005) recommend that teachers use technology when it can “enhance and enrich their performances of understanding. . . . Word processors, digital audio and video technologies for creating Web sites allow students to express their understanding in a rich variety of media. These technologies also capture student work in forms that can be easily revised, combined, and distributed” (Wiske et al., 2005, pp. 65–66).
4. *Consider how students and teachers assess learning.* Use ongoing assessment instead of only using a final assessment. During ongoing assessment, you might guide students in understanding what quality work involves or use peer collaboration to help students analyze and improve their work. A helpful strategy is to also encourage students to assess their own learning progress and to monitor how effectively they are learning. Technology can be used in several ways to effectively assess learning.

“Digital technologies, including audio and video recorders and computers, can capture student work in forms that are easy to review. Interactive workspaces and software with multiple windows can help to keep assessment guidelines in view and even offer prompts and reminders. . . . Using networked technologies, students may post their work online where it can be readily reviewed and annotated by multiple advisors, including distant teachers and peers who cannot meet face-to-face. Technologies also provide easy means of preserving digital archives of student work. These may allow teachers and students to create individual portfolios to demonstrate and evaluate a student’s work over time” (Wiske et al., 2005, pp. 84–85).



Students working on a GenYes project. What are some characteristics of GenYes?

Generation YES

5. *Reflect on how students and teachers can learn together.* “Networked technologies provide multiple advantages for connecting learners with reflective, collaborative communities. . . .” For example, “E-mail permits users to send and receive many-to-many messages and to do so quickly. Students can share information and work with many other students all over the world, exchanging multiple rounds of reflective dialogue. The Web, with digital images, video and audio recordings, and videoconferencing, also allows students and their teachers to publish and collaborate on work, opening up the possibility of communicating with a wide range of audiences outside the classroom” (Wiske et al., 2005, pp. 100, 102). Social media also can be an important tool to accomplish these goals as well.

In further considering the role of technology in learning, a technology program that is increasingly used in schools is the GenYes program. The goal of the GenYes program is to support the effective integration of technology in teaching and learning (GenYes, 2010). The program, created by Dennis Harper at the Northwest Regional Educational Laboratory, emphasizes that teachers and students are partners in creating lessons that use technology in ways students find meaningful and relevant. GenYes is available for both elementary and secondary school students, who learn by co-creating lessons with teachers while their teachers learn about technology from the students. Instead of teaching technology skills to teachers with the hope that they will use those skills in teaching students, GenYes works with students to help them form productive technology partnerships with their teachers. Students and their teacher-partners learn how to use telecommunication devices, the Internet, presentation tools, and other emerging technologies.

Among the technology units that GenYes students and teachers collaborate on are:

- *Online communications*, such as safety, e-mail, forums, blogs, messaging, and collaborative projects
- *Digital media*, such as graphics, animation, and video
- *Digital authoring*, including planning, creating, and delivering multimedia projects
- *Web publishing*, such as designing and constructing Web pages
- *Student leadership and community service*

In addition, a number of mini-units, such as podcasting, digital storytelling, handhelds, and video tutorials can be incorporated in the GenYes class or used by students when they are working on technology projects.

GenYes students are paired, either individually or in teams, with a partner-teacher. Initial meetings are held to determine a curriculum focus that could be enriched by an infusion of technology. The GenYes students are responsible for the nuts and bolts of the technology, and the teacher provides content accuracy and pedagogical strategies.

The GenYes program was initiated in Olympia, Washington, in 1996 and is now used in a number of classrooms across the United States. Teachers and students have consistently reported that GenYes has provided them with an excellent opportunity to improve their technology skills. The GenYes program reflects the learner-centered strategies we discussed earlier in this chapter and the social constructivist approaches described in the chapter on that topic.

Technological Pedagogical Content Knowledge (TPCK) *Technological Pedagogical Content Knowledge (TPCK)* is a model created by Matthew Koehler and Punya Mishra (2010; Mishra & Koehler, 2006) that emphasizes the importance of not looking at technology as a stand-alone entity but rather addressing links between technology, content knowledge, and pedagogy (see Figure 5). In their view, knowing how to use

technology is not sufficient for instructional success in the classroom; you also need to know how to teach with it. They argue that teachers who are skilled in all three domains—technology, content knowledge, and pedagogy—have a level of expertise in teaching a particular topic using technology that differs from that of a technology expert (e.g., a computer scientist), a content knowledge expert in a particular discipline (e.g., science), or a pedagogical expert (e.g., an experienced educator). TPACK has been adopted by the American Association of Colleges of Teacher Education, which sponsored a book by Koehler and Mishra (2010) on the integration of technology, content knowledge, and pedagogy to improve classroom instruction. Educators are showing continuing interest in using TPACK in their classroom teaching (Kennedy, 2015; Niess & Gillow-Wiles, 2014).

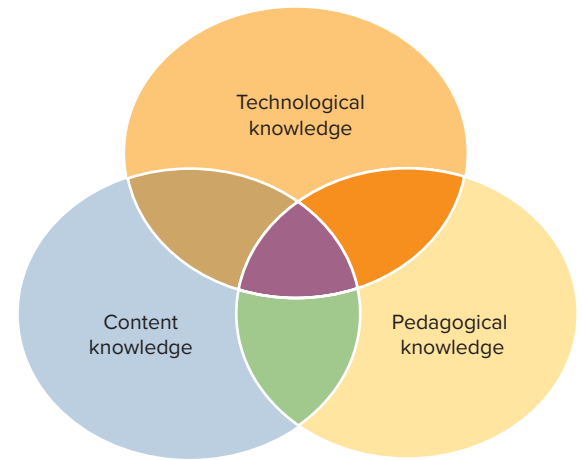


FIGURE 5 TECHNOLOGICAL PEDAGOGICAL CONTENT KNOWLEDGE

Stages of Integrating Technology into Classroom Teaching Integrating technology into your classroom teaching often takes place in the following sequence (Norris & Soloway, 1999):

- Stage 1: Teacher is aware that a particular technology exists but hasn't used it; may be avoiding the technology.
- Stage 2: Teacher is currently trying to learn the basic aspects of the technology but often becomes frustrated and still lacks confidence with this technology.
- Stage 3: Teacher is beginning to see how to use a particular technology and thinks about specific situations in which to use it.
- Stage 4: Teacher is gaining confidence in using the technology for certain tasks and feeling more comfortable in using it.
- Stage 5: Teacher now thinks about this technology as something that helps him or her and no longer lacks confidence in using it; teacher perceives that the technology can be used in a number of instructional contexts.
- Stage 6: Teacher can use this technology as an effective instructional tool across the curriculum to meet instructional objectives.

To further evaluate your technology skills and attitudes, complete *Self-Assessment 1*.

How do teachers use technology to help students learn? Following are their responses.

EARLY CHILDHOOD Technology helps preschoolers learn music—in addition to listening to music to learn a song, they can also see the written words projected on a screen along with appropriate clip art.



—CONNIE CHRISTY, *Aynor Elementary School (Preschool Program)*

ELEMENTARY SCHOOL: GRADES K-5 I often have my fifth-grade students research topics using Google and have them sift through the information. We discuss valid sources and try to determine ones that are questionable.



—CRAIG JENSEN, *Cooper Mountain Elementary School*

MIDDLE SCHOOL: GRADES 6-8 As a Spanish language teacher, I devised a plan with a teacher from Chile in which my students e-mailed her students in Spanish. In turn, her students e-mailed mine in English. This bilingual process helped both groups of students learn the respective languages of each country.



—MARGARET REARDON, *Pocantico Hills School*



How good are your technology skills? How positive are your attitudes about using technology and incorporating it into your classroom? For these items, consider the grade and subject(s) you are most likely to teach. Rate yourself from 1 to 5, with 1 = Not like me at all and 5 = Very much like me.

[illegible]

1. I'm reasonably proficient at using a computer, including installing and uninstalling software.
2. I am able to use tablets and mobile devices with relative ease.
3. I know when and how to use technology to improve students' understanding.
4. I have ideas for using word processing together with other language learning resources in the classroom and know the best apps to use in teaching.
5. I know how to search efficiently and thoroughly for information that interests me on the Internet.
6. I can vet the quality, accuracy, and sufficiency of information that I find on the Internet.
7. I have ideas about how to use the Internet in my classroom.
8. I'm proficient at using e-mail.
9. I know how to use PowerPoint.
10. I have been part of collaborative learning exercises that involve technology.
11. I can see how collaborative learning can be used with technology in my classroom.
12. I am aware of the sociocultural issues involved in technology and education.
13. I know some good Web sites, journals, and software catalogs that can help me learn how to use technology more effectively in the classroom.
14. I have a good understanding of social media as a tool for communication.
15. I have a good understanding of how to use social media (e.g., blogs, Instagram, Pinterest) for professional development.
16. I know what cloud computing is and how to use it.
17. I understand online safety and etiquette.
18. I know how to use online tools for assessment and student tracking.

Look at your scores for each item and evaluate your technology strengths and weaknesses. By the time you step into your classroom for your first day of teaching, make it a goal to be able to confidently rate yourself on each of these items at the level of 4 or 5. On items on which you rated yourself 1, 2, and 3, try to take technology courses at your local college that will improve your knowledge and skills in those areas.

—JENNIFER HEITER, *Bremen High School*



CONNECTING WITH STUDENTS: Best Practices

Strategies for Choosing and Using Technology in the Classroom

Technology will be a part of your classroom. Here are some guidelines for choosing and using it:

1. *Choose technology with an eye toward how it can help students actively explore, construct, and restructure information.* Look for software, apps, and online (cloud-based) tools and sites that let students directly manipulate the information. One review found that students' learning improved when information was presented in a multimedia fashion that stimulated them to actively select, organize, and integrate visual and verbal information (Mayer, 1997). You might want to consult with a school or district media specialist to find the software that best reflects these characteristics. An excellent resource for teachers that emphasize the use of technology to improve students' learning and understanding is the International Society for Technology in Education (ISTE) Web site (www.iste.org). Software catalogs and journals also are good sources of information.
2. *Look for ways to use technology as part of collaborative and real-world learning.* In Ann Brown and Joe Campione's (1996) words, education should be about "fostering a community of learners." Students often learn better when they work together to solve challenging problems and construct innovative projects (Kaendler et al., 2015). The term *computer-supported collaborative learning* (CSCL) has been coined to describe learning that occurs when students share and construct knowledge while interacting with others using a computer or the Internet (Kirschner et al., 2015; Rodriguez-Triana et al., 2015; Xiong et al., 2015). One example of CSCL is collaborative writing, which motivates students to generate ideas and share them with others to develop a better understanding of a topic. This might be accomplished using blogs, interactive
- whiteboards, and other communication tools. Also, think of technologies such as the Web, e-mail, social media, and the Cloud as tools for providing students with opportunities to engage in collaborative learning. In this way, students can reach outside the classroom to include the real world and communicate with people in locations that otherwise would be inaccessible to them.
3. *Choose technology that presents positive models for students.* Monitor technology for equity in ethnicity and culture, and be sure that the models presented are positive role models.
4. *Remember that your teaching skills are critical, regardless of the technology you use.* You don't have to worry that technology will replace you as a teacher. Technology becomes effective in the classroom only when you know how to use it, demonstrate it, guide and monitor its use, and incorporate it into a larger effort to develop students who are motivated to learn, actively learn, and communicate effectively. Even the most sophisticated digital tools will not benefit students much unless you appropriately orient students to it, ask them good questions about the material, orchestrate its use, and tailor it to their needs.
5. *Continue to learn about technology yourself and increase your technological competence.* Digital technology is still changing at an amazing pace. Make it a personal goal to be open to new technology, keep up with technological advances by reading educational journals, and take courses in educational computing to increase your skills. You will be an important model for your students in terms of your attitude toward technology, your ability to use it effectively yourself, and your ability to communicate how to use it effectively to your students.



Review, Reflect, and Practice

- 4 Summarize how to effectively use technology to help children learn.

REVIEW

- What characterizes the technology revolution and the Internet?
- What are some standards for technology-literate students?
- What are five things that teachers need to ask when considering the use of technology in the classroom to improve students' understanding?

REFLECT

- Would one or more of the ways that computers can be used to support learning and instruction be useful to the subject and grade level you plan to teach? How?

continued

Review, Reflect, and Practice

PRAXIS™ PRACTICE

- Ms. Carlson's students are working on assignments posted onto their school server through the Internet. This delivery system is referred to as
 - in-house intranet
 - social media for kids
 - cloud computing
 - Web enhancement
- Mr. Wilson's social studies students are examining various websites to research regional population growth and urban planning around the world. This activity is best aimed at which grade levels?
 - PK–2
 - 3–5
 - 6–8
 - 9–12
- Which of the following examples best reflects the effective use of technology to promote student understanding?
 - Roberto is playing an online multiplication facts game that provides points for correct answers and feedback indicating the correct answer when he is wrong.
 - Patricia is using a word-processing program to type a paper for her English class after having written it out longhand.
 - Deshawn is immersed in a technology simulation of the desert ecosystem.
 - Carmine is texting his best friend during class regarding their plans for the evening.

Please see answer key at end of chapter

Connecting with the Classroom: Crack the Case

The Big Debate

Mrs. Rumer was new to teaching third grade at Hillside Elementary School. Before the school year began, she met with other new teachers and their mentors for planning sessions. The administration appeared to be aware of just how much planning was necessary for successful teaching. Mrs. Rumer openly shared her ideas with her mentor, Mrs. Humbolt, and the rest of the group.

"I really want to have a learner-centered classroom," she said. "I'd like to use aspects of problem-based learning, essential questions, and guided discovery. I think the students will learn so much more that way than if I use teacher-centered instruction."

Mrs. Humbolt smiled and said, "Well, they'd probably have more fun, but I doubt that their test scores would reflect much learning at all. We really need to prepare our students to meet state standards, Mrs. Rumer. To do that, you'd better throw in some good old-fashioned direct instruction." Several other teachers readily agreed. One commented, "That constructivist stuff is too much fluff; I want my students to be serious and learn what I teach them." Another said, "I use the computers in the classroom for

drilling students to memorize material for the state tests they have to take, sort of like giving them electronic flash cards. I guess that wouldn't fit your scheme."

The other teachers' comments surprised Mrs. Rumer. She had learned in her education courses that learner-centered instruction is supposed to be the best way to teach children. She wanted her students to actively construct their knowledge, not merely have her pour information into their minds. The principal assured her that if she wanted to use a learner-centered approach, she would have that freedom.

With this assurance, Mrs. Rumer began making lists of everything she would have to plan for in order to have an effective learner-centered classroom. She began by going through the district's curriculum guide for third grade. She made lists of all the objectives. Then she went through the learner-centered psychological principles from the APA. After doing this, she realized her job was going to be a daunting one.



1. What are the issues in this case?
2. Where should Mrs. Rumer go from here?
3. How can she take a curriculum that has been taught in a teacher-centered manner and convert it to a learner-centered curriculum? Should she? Why or why not?
4. How can she incorporate technology into the curriculum so that the computers don't become mere electronic flashcards?
5. Which of the following is an activity that would most likely appeal to Mrs. Rumer?
 - a. The students will master their basic multiplication facts by completing a worksheet covering them each day.
 - b. The students will master their basic multiplication facts by playing multiplication baseball.
 - c. The students will master their basic multiplication facts by taking daily timed tests covering them.
 - d. The students will master their basic multiplication facts by writing out their multiplication tables repeatedly.
6. Which of the following is an activity that would most likely appeal to Mrs. Rumer's colleagues?
 - a. The students will learn the scientific process by completing several experiments.
 - b. The students will learn the scientific process by focusing on essential questions in science.
 - c. The students will learn the scientific process by reading about it in their science texts and listening to lectures.
 - d. The students will learn the scientific process by testing the water in a nearby creek.

Connecting with Learning: Reach Your Learning Goals

1 PLANNING: Explain what is involved in classroom planning.

Instructional Planning

- Instructional planning involves developing a systematic, organized strategy for lessons. Planning is critically important to being a competent teacher, and instructional planning may be mandated by schools.

Time Frames and Planning

- Teachers need to make plans for different time frames, ranging from yearly planning to daily planning.
- Yinger identified five time frames of teacher planning: yearly, term, unit, weekly, and daily.

2 TEACHER-CENTERED LESSON PLANNING AND INSTRUCTION: Identify important forms of teacher-centered instruction.

Teacher-Centered Lesson Planning

- Teacher-centered lesson planning includes creating behavioral objectives, analyzing tasks, and developing instructional taxonomies (classifications).
- Behavioral objectives are statements that propose changes in students' behavior to reach desired performance levels.
- Task analysis focuses on breaking down a complex task that students are to learn into its component parts.
- Bloom's taxonomy classifies educational objectives into cognitive, affective, and psychomotor domains and is used by many teachers to create goals and objectives in lesson planning.

Direct Instruction

- Direct instruction is a structured, teacher-centered approach that involves teacher direction and control, high expectations for students' progress, maximum time spent by students on academic tasks, and efforts by the teacher to keep negative affect to a minimum. The teacher chooses students' learning tasks and directs students' learning of those tasks. The use of nonacademic materials is deemphasized, as is nonacademically oriented teacher-student interaction.

Teacher-Centered Instructional Strategies

- Teacher-centered instructional strategies include orienting students to new material; lecturing, explaining, and demonstrating; questioning and discussing; mastery learning (learning one topic or concept thoroughly before moving on to a more difficult one); seatwork (having students work independently at their seats); and homework.

continued

Evaluating Teacher-Centered Instruction

- Teacher-centered instruction includes useful techniques, and its advocates especially believe it is the best strategy for teaching basic skills such as math computations, grammar rules, and reading vocabulary. Critics of teacher-centered instruction say that by itself it tends to lead to passive rote learning, overly rigid and structured classrooms, inadequate attention to socioemotional development, external rather than internal motivation, excessive use of paper-and-pencil tasks, too few opportunities for real-world learning, and too little collaborative learning in small groups.

3 LEARNER-CENTERED LESSON PLANNING AND INSTRUCTION: Discuss important forms of learner-centered instruction.

Learner-Centered Principles

- Learner-centered lesson planning and instruction move the focus away from the teacher and toward the student.
- The APA's learner-centered psychological principles emphasize the active, reflective nature of learning and learners. The fourteen principles involve cognitive and metacognitive factors (the nature of the learning process, goals of the learning process, the construction of knowledge, strategic thinking, thinking about thinking, and the context of learning); motivational and emotional factors (motivational and emotional influences on learning, intrinsic motivation to learn, and effects of motivation on effort); developmental and social factors (developmental influences on learning and social influences on learning); and individual difference factors (individual differences in learning, learning and diversity, and standards and assessments).

Some Learner-Centered Instructional Strategies

- Problem-based learning emphasizes real-world problem solving. A problem-based curriculum exposes students to authentic problems.
- Problem-based learning focuses on small-group discussion rather than lecture. Students identify issues they wish to explore, and teachers act as guides, helping students monitor their problem-solving efforts.
- Essential questions are questions that engagingly reflect the heart of the curriculum, cause students to think, and motivate their curiosity.
- Discovery learning is learning in which students construct an understanding on their own. Most discovery-learning approaches today involve guided discovery, in which students are encouraged to construct their understanding with the assistance of teacher-guided questions and directions.

Evaluating Learner-Centered Strategies

- The learner-centered model of planning and instruction has many positive features. The fourteen APA learner-centered principles are guidelines that can help teachers develop strategies that benefit student learning (such as encouraging students to actively construct knowledge, think deeply and creatively, be internally motivated, solve real-world problems, and collaboratively learn).
- Critics argue that learner-centered planning and instruction focuses too much on the process of learning and not enough on academic content; is more appropriate for social sciences and humanities than science and math; is not appropriate for beginning instruction when students have little or no knowledge about the topic; and is more challenging to implement than most teachers envision.
- Keep in mind that although we presented teacher-centered and learner-centered approaches separately, many teachers use aspects of both approaches.

4 TECHNOLOGY AND EDUCATION: Summarize how to effectively use technology to help children learn.

The Technology Revolution and the Internet

- The technology revolution is part of the information society in which we now live, and students will increasingly need to have technological skills. Today's technologies can be remarkable tools for motivating students and guiding their learning.
- The Internet is the core of computer-mediated communication. The Web is a system for browsing Internet sites. The Internet can be an important learning tool in many classrooms. Cautions about Internet use need to be observed.



- The International Society for Technology in Education (ISTE) has established technology standards for students that include using digital technologies to support being a(an) empowered learner, digital citizen, knowledge constructor, innovative designer, computational thinker, creative communicator, and global collaborator. In addition, ISTE provides performance indicators for achieving these standards at different grade levels.
- Teachers need to consider five things when exploring how to use technology more effectively in the classroom to improve students' learning: (1) which topics are worth understanding, (2) what students should understand about such topics, (3) how students develop and demonstrate understanding, (4) how students and teachers assess understanding, and (5) how students and teachers learn together.

KEY TERMS

advance organizers 402
behavioral
objectives 398
Bloom's taxonomy 398
cloud computing 416

comparative advance
organizers 402
direct instruction 400
discovery learning 411
essential questions 411

expository advance
organizers 402
guided discovery learning 411
instructional planning 394
Internet 416

mastery learning 404
task analysis 398
taxonomy 398
Web 416

PORTFOLIO ACTIVITIES

Now that you have a good understanding of this chapter, complete these exercises to expand your thinking.

Independent Reflection

1. **Developing a Classroom Technology Plan.** Create a written plan for how you might use digital technologies for the subject(s) and grade level you plan to teach. How will you adapt your plan for students with little or no experience with digital technologies? How can your classroom benefit from students with advanced technology skills? (INTASC: Principles 1, 4, 8)

Research/Field Experience

2. **Instructional Planning in Action.** Ask a teacher at the grade level you plan to teach to show you the materials he or she uses in

planning lessons, units, the term, and the yearly curriculum for one or more subjects. Create samples for your own later use based on what the teacher shows you. Discuss the importance of planning at each of these levels. (INTASC: Principles 7, 9)

Collaborative Work

3. **Evaluating Teacher-Centered and Learner-Centered Classrooms.** With three other students in the class, divide up the work of observing an early childhood, an elementary, a middle school, and a high school classroom. Reconvene after each of you has observed a classroom, and discuss the aspects of teacher-centered and learner-centered approaches the teachers were using. Evaluate how effective the approaches were. Write a comparative analysis. (INTASC: Principles 1, 2, 3, 4, 5, 6, 7, 8, 9)

PRAXIS™ PRACTICE ANSWER KEY

Planning

1. b
2. d

Teacher-Centered Lesson Planning and Instruction

1. a 3. a
2. c 4. c

Learner-Centered Lesson Planning and Instruction

1. d
2. c
3. b

Technology and Education

1. c
2. d
3. a